CS-1101
PROGRAMMING FUNDAMENTALS

Lab 08

By

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Lab Contents

- Control Structures
- Decision Control Structures
  - switch

Lab Objectives

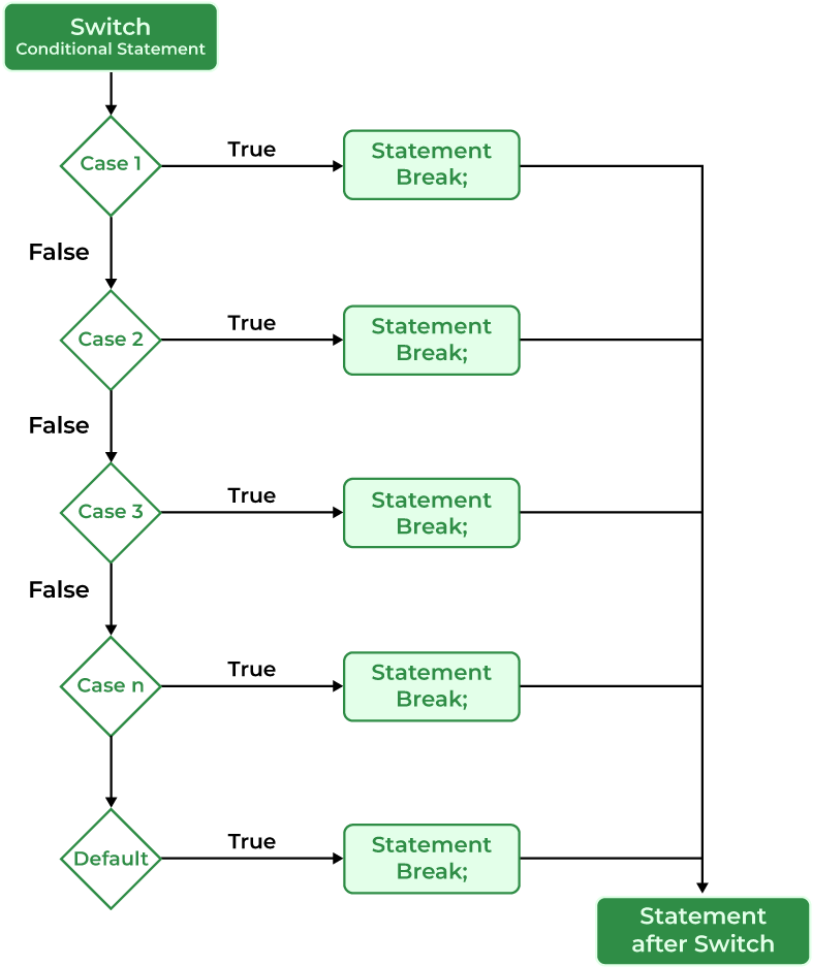
- The objective of this lab is to learn how to use switch statements to control the flow of a program based on specific user inputs.

Switch

- Switch statement in C++ is a flow control statement that is used to execute the different blocks of statements based on the value of the given expression.

```
switch (expression) {
    case value_1:
        // statements_1;
        break;
    case value_2:
        // statements_2;
        break;
    .....
    .....
    default:
        // default_statements;
        break;
}
```

Flowchart of Switch Statement in C++



Example of Switch

```
#include<iostream>
using namespace std;
int main(){
    char x = 'A';
    // switch statements
    switch (x) {
    case 'A':
        cout << "Choice is A";
        break;
    case 'B':
        cout << "Choice is B";
        break;
    case 'C':
        cout << "Choice is C";
        break;
    default:
        cout << "Choice other than A, B and C";
        break;
    }
    return 0;
}
```

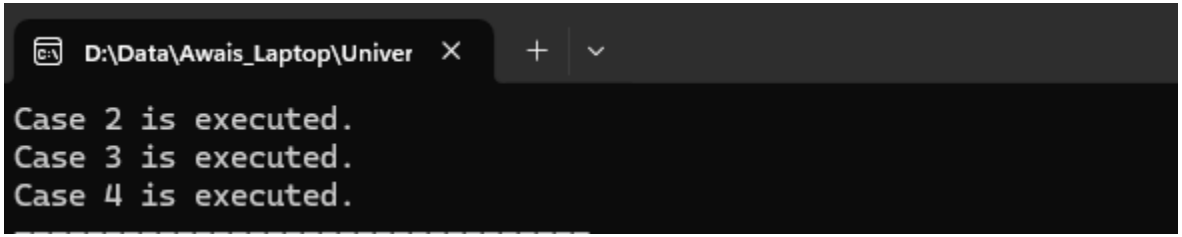
```
Choice is A
```

Rules of the switch case statement in C++

- There are some rules that we need to follow when using switch statements in C++. They are as follows:
  - The switch statement can only evaluate the integer or character value. So, the switch expression should return the values of type int or char only.
  - The case value must be either int or char type.
  - There can be any number of cases.
  - No duplicate case values are allowed.
  - Each statement of the case can have a break statement. It is optional.
  - The default statement is also optional.
  - Position of Default Case Does Not Matters.

Example of switch cases without the break

```
#include<iostream>
using namespace std;
int main(){
    int var = 2;
    // switch case without break
    switch (var) {
    case 1:
        printf("Case 1 is executed.\n");
    case 2:
        printf("Case 2 is executed.\n");
    case 3:
        printf("Case 3 is executed.\n");
    case 4:
        printf("Case 4 is executed.");
    }
    return 0;
}
```



Tasks – switch

- Write a C++ program that acts as a simple calculator. The program should take two numbers and an operator (+, -, *, /) as inputs and perform the corresponding operation using a switch statement. If an invalid operator is entered, the program should display an appropriate error message.
- Write a C++ program that takes an integer (1-7) as input and uses a switch statement to display the corresponding day of the week (e.g., 1 for Monday, 2 for Tuesday, etc.). If the number is not in the range of 1 to 7, display an error message.
- Write a C++ program that calculates the area of different shapes. Use a switch statement to let the user select a shape (1 for Circle, 2 for Rectangle, 3 for Triangle) and input the necessary dimensions. The program should compute and display the area based on the selected shape.

Tasks – switch

- Write a C++ program that acts as a currency converter. Use a switch statement to let the user choose between converting from USD to PKR,INR,Euros. Display the converted amount based on the exchange rate and the amount entered by the user.

Shopping System with Invoice Generation and Discounts

- A customer visits a shopping system where they can choose from three categories of items: Electronics, Clothing, and Groceries. Each category has multiple items with specified prices. The customer selects a category and then select an item with in the category, specifies the quantity they wish to purchase, and the system calculates the total price, applies applicable discounts, and generates an invoice.
- Discount Policy:
  - Purchases below $100: No discount.
  - Purchases between $100 and $500: 10% discount.
  - Purchases above $500: 20% discount.
- Items & Prices:

  The prices for items are predefined:
  - **Electronics:** Laptop ($1000), Smartphone ($700), Headphones ($150)
  - **Clothing:** Jacket ($120), T-shirt ($40), Jeans ($60)
  - **Groceries:** Milk (1 Liter - $2), Bread (1 Loaf - $3), Eggs (1 Dozen - $5)

References

- https://www.geeksforgeeks.org/switch-statement-in-cpp/